ORTO-DIAGONALTZAREA MATRICELOR SIMETRICE

Valori proprii + Vectori proprii + Gram-Schmidt

1. Algoritm pentru ortodiagonalizarea matricelor sim.:

Teorema de descompunere spectrală

Teoremă 7.51. Fie $A \in \mathcal{M}_n(\mathbb{R})$ o matrice. Atunci A este **simetrică** dacă şi numai dacă există o matrice **ortogonală** $Q \in \mathcal{M}_n(\mathbb{R})$ şi o matrice **diagonală** $D \in \mathcal{M}_n(\mathbb{R})$ astfel încât

$$A = QDQ^t$$

În acest caz, mai spunem că A este **orto-diagonalizabilă**.

a) $A = \begin{pmatrix} 4 & 2 \\ 2 & 7 \end{pmatrix}$

Determinăm valorile proprii

$\det(A - \lambda I_2) = (4-\lambda)(7-\lambda) - 4 = 28 - 4\lambda - 7\lambda + \lambda^2 - 4 =$

$= \lambda^2 - 11\lambda + 24$

$\Delta = 121 - 4 \cdot 24 = 121 - 96 = 25$

$\lambda_{1,2} = \frac{11 \pm 5}{2}$ ③ ⑧

Pentru $\lambda_1 = 3 \Rightarrow$

$\begin{cases} 4x + 2y = 3x \\ 2x + 7y = 3y \end{cases} \Leftrightarrow \boxed{x = -2y} \Rightarrow V_{\lambda_1} = Sp\{\begin{pmatrix} -2 \\ 1 \end{pmatrix}\}$

Pentru $\lambda_2 = 8 \Rightarrow$

$\begin{cases} 4x + 2y = 8x \\ 2x + 7y = 8y \end{cases} \Rightarrow 2y = 4x \Rightarrow y = 2x \Rightarrow V_{\lambda_2} = Sp\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\}$

$v_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Aplicăm algoritmul Gram-Schmidt.

$u_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ $q_1 = \frac{1}{\|u_1\|} \cdot u_1 = \frac{1}{\sqrt{2^2+1^2}} \cdot u = \begin{pmatrix} \frac{-2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$

$u_2 = v_2 - \overbrace{pr_{q_1} v_2}^{\neq 0} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{\langle v_2, q_1 \rangle}{\langle q_1, q_1 \rangle} \cdot q_1 =$

$= \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \frac{\frac{-2}{\sqrt{5}} + \frac{2}{\sqrt{5}}}{1} \cdot q_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$q_2 = \frac{1}{\|u_2\|} \cdot u_2 = \frac{1}{\sqrt{5}} \cdot u_2 = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix} \Rightarrow$

$A = \underbrace{\begin{pmatrix} \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}}_{\text{o matrice ortogonală}} \underbrace{\begin{pmatrix} 3 & 0 \\ 0 & 8 \end{pmatrix}}_{\text{o matrice diagonală}} \begin{pmatrix} \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}^t$

(b) $A = \begin{pmatrix} 2 & 3 & 0 \\ 3 & 2 & 4 \\ 0 & 4 & 2 \end{pmatrix}$ $A - \lambda I_3 = \begin{pmatrix} 2-\lambda & 3 & 0 \\ 3 & 2-\lambda & 4 \\ 0 & 4 & 2-\lambda \end{pmatrix}$

$\det(A - \lambda I_3) = -\lambda^3 + 6\lambda^2 + 13\lambda - 42 = (\lambda-2)(\lambda+3)(\lambda-7)$

Pentru $\lambda = 2 \Rightarrow$

$\begin{cases} 2x+3y = 2x \\ 3x+2y+4z = 2y \\ 4y+2z = 2z \end{cases} \Leftrightarrow \begin{cases} y = 0 \\ 3x = -4z \end{cases} \Rightarrow V_{\lambda_2} = Sp\left\{ \begin{pmatrix} -4 \\ 0 \\ 3 \end{pmatrix} \right\}$

$v_2 = \begin{pmatrix} -4 \\ 0 \\ 3 \end{pmatrix}$

Pentru $\lambda = -3 \Rightarrow$

$\begin{cases} 2x+3y = -3x \\ 3x+2y+4z = -3y \\ 4y+2z = -3z \end{cases} \Leftrightarrow \begin{cases} -5x = 3y \\ 4y = -5z \end{cases} \Leftrightarrow \begin{cases} x = \frac{-3}{5}y \\ z = \frac{-4}{5}y \end{cases} \Rightarrow$

$V_{\lambda_1} = Sp\left\{ \begin{pmatrix} -3/5 \\ 1 \\ -4/5 \end{pmatrix} \right\} = Sp\left\{ \begin{pmatrix} 3 \\ -5 \\ 4 \end{pmatrix} \right\}, v_1 = \begin{pmatrix} 3 \\ 5 \\ -4 \end{pmatrix}$

Pentru $\lambda = 7 \Rightarrow$

$$\begin{cases} 2x+3y=7x \\ 3x+2y+4z=7y \\ 4y+2z=7z \end{cases} \Leftrightarrow \begin{cases} 3y=5x \\ 4y=5z \end{cases} \Rightarrow \begin{cases} x=\frac{3}{5}y \\ z=\frac{4}{5}y \end{cases} \Rightarrow V_{12}=\mathcal{S}p\left\{\begin{pmatrix}3\\5\\4\end{pmatrix}\right\}$$

$$v_3=\begin{pmatrix}3\\5\\4\end{pmatrix}$$

Aplicăm algoritmul Gram-Schmidt

$u_1 = v_1 \quad q_1 = \frac{1}{\|u_1\|}\cdot u_1 = \frac{1}{\sqrt{9+25+16}}\cdot u_1 = \frac{1}{5\sqrt{2}}\cdot u_1 = \begin{pmatrix}\frac{-3}{5\sqrt{2}}\\ \frac{1}{\sqrt{2}}\\ \frac{-4}{5\sqrt{2}}\end{pmatrix}$

$u_2 = v_2 - \underbrace{pr_{q_1} v_2}_{\|\cdot\|=1\ (*)} = v_2$

$q_2 = \frac{1}{\|u_2\|}\cdot v_2 = \frac{1}{\sqrt{16+9}} = \begin{pmatrix}\frac{-4}{5}\\0\\\frac{3}{5}\end{pmatrix}$

$u_3 = v_3 - \underbrace{pr_{q_1} v_3}_{=0\ (*)} - \underbrace{pr_{q_2} v_3}_{=0\ (*)} = v_3$

$q_3 = \frac{1}{\|u_3\|}\cdot u_3 = \frac{1}{5\sqrt{2}}\cdot\begin{pmatrix}3\\5\\4\end{pmatrix} = \begin{pmatrix}\frac{3}{5\sqrt{2}}\\ \frac{1}{\sqrt{2}}\\ \frac{4}{5\sqrt{2}}\end{pmatrix} \Rightarrow$

$Q=(q_1|q_2|q_3)$

$D=\begin{pmatrix}-3&0&0\\0&2&0\\0&0&7\end{pmatrix}$ $\Rightarrow A=\underbrace{\begin{pmatrix}\frac{-3}{5\sqrt{2}}&\frac{-4}{5}&\frac{3}{5\sqrt{2}}\\ \frac{1}{\sqrt{2}}&0&\frac{1}{\sqrt{2}}\\ \frac{-4}{5\sqrt{2}}&\frac{3}{5}&\frac{4}{5\sqrt{2}}\end{pmatrix}}_{Q}\begin{pmatrix}-3&0&0\\0&2&0\\0&0&7\end{pmatrix}\cdot Q^t$

Lemă 7.50. Fie $A \in \mathcal{M}_n(\mathbb{R})$ o matrice simetrică.
Dacă $\lambda_1, \lambda_2 \in \mathbb{R}$ sunt valori proprii **distincte** ale matricei A şi $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$ sunt vectori proprii asociaţi valorilor proprii λ_1, λ_2, atunci $\mathbf{v}_1, \mathbf{v}_2$ sunt vectori **ortogonali**.

↳ conform acestei leme proiecțiile din algoritmul Gram-Schmidt sunt egale cu zero

(c) $A=\begin{pmatrix}1&0&-1\\0&1&0\\-1&0&1\end{pmatrix}$ $A-\lambda I_3=\begin{pmatrix}1-\lambda&0&-1\\0&1-\lambda&0\\-1&0&1-\lambda\end{pmatrix}$

$\det(A-\lambda I_3)=-\lambda^3+3\lambda^2-2\lambda=-\lambda(\lambda^2-3\lambda+2)=-\lambda(\lambda-1)(\lambda-2)$

Pentru $\lambda_1=0\Rightarrow$

$\begin{cases}x-z=0\\y=0\\-x+z=0\end{cases}\Rightarrow\begin{cases}x=z\\y=0\end{cases}\Rightarrow V_{\lambda_1}=Sp\left\{\begin{pmatrix}1\\0\\1\end{pmatrix}\right\}, v_1=\begin{pmatrix}1\\0\\1\end{pmatrix}$

Pentru $\lambda_2=1\Rightarrow$

$\begin{cases}x-z=x\\y=y\\-x+z=z\end{cases}\Rightarrow\begin{cases}x=z=0\\y\in\mathbb{R}\end{cases}\Rightarrow V_{\lambda_2}=Sp\left\{\begin{pmatrix}0\\1\\0\end{pmatrix}\right\}, v_2=\begin{pmatrix}0\\1\\0\end{pmatrix}$

Pentru $\lambda_3=2\Rightarrow$

$\begin{cases}x-z=2x\\y=2y\\-x+z=2z\end{cases}\Rightarrow\begin{cases}x=-z\\y=0\end{cases}\Rightarrow V_{\lambda_3}=Sp\left\{\begin{pmatrix}1\\0\\-1\end{pmatrix}\right\}, v_3=\begin{pmatrix}1\\0\\-1\end{pmatrix}$

Am aplicat direct lema anterioară

$q_1=\frac{1}{\|v_1\|}\cdot v_1=\frac{1}{\sqrt2}\cdot\begin{pmatrix}1\\0\\1\end{pmatrix}=\begin{pmatrix}1/\sqrt2\\0\\1/\sqrt2\end{pmatrix}$ $q_3=\frac{1}{\|v_3\|}\cdot v_3=\begin{pmatrix}1/\sqrt2\\0\\-1/\sqrt2\end{pmatrix}$

$q_2=\frac{1}{\|v_2\|}\cdot v_2=\begin{pmatrix}0\\1\\0\end{pmatrix}$ $Q=(q_1|q_2|q_3)$ $D=\begin{pmatrix}\lambda_1&0&0\\0&\lambda_2&\\0&&\lambda_3\end{pmatrix}$

$\Rightarrow A=QDQ^t=\begin{pmatrix}1/\sqrt2&0&1/\sqrt2\\0&1&0\\1/\sqrt2&0&-1/\sqrt2\end{pmatrix}\begin{pmatrix}0&0&\\0&1&\\&&2\end{pmatrix}\begin{pmatrix}1/\sqrt2&0&1/\sqrt2\\0&1&0\\1/\sqrt2&0&-1/\sqrt2\end{pmatrix}^t$

(d) $A=\begin{pmatrix}1&1&1\\1&1&1\\1&1&1\end{pmatrix}$ $A-\lambda I_3=\begin{pmatrix}1-\lambda&1&1\\1&1-\lambda&1\\1&1&1-\lambda\end{pmatrix}$

$\det(A-\lambda I_3)=-\lambda^3+3\lambda^2=-\lambda^2(\lambda-3)$

Pentru $\lambda=0\Rightarrow$

$x+y+z=0\Rightarrow z=-x-y\Rightarrow V_{\lambda_1}=Sp\left\{\underbrace{\begin{pmatrix}1\\0\\-1\end{pmatrix}}_{v_1},\underbrace{\begin{pmatrix}0\\1\\-1\end{pmatrix}}_{v_2}\right\}$

Pentru $\lambda=3\Rightarrow$

$\begin{cases}x+y+z=3x\\x+y+z=3y\\x+y+z=3z\end{cases}\Rightarrow x=y=z\Rightarrow V_{\lambda_2}=Sp\left\{\underbrace{\begin{pmatrix}1\\1\\1\end{pmatrix}}_{v_3}\right\}\Rightarrow$

Avem o valoare proprie cu $m_a=2$ deci v_1 și v_2 nu sunt ortogonali, deci aplicăm alg. G-S.

$u_1 = v_1 \quad q_1 = \frac{1}{\|u_1\|} \cdot u_1 = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$

$u_2 = v_2 - pr_{q_1} v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} - \frac{\langle v_2, q_1 \rangle}{\langle q_1, q_1 \rangle} \cdot q_1 = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix} -$

$\frac{1/\sqrt{2}}{1} \cdot \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1/2 \\ 0 \\ -1/2 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1 \\ -1/2 \end{pmatrix}$

$q_2 = \frac{1}{\|u_2\|} \cdot u_2 = \frac{1}{\sqrt{1 + \frac{1}{2}}} \cdot \begin{pmatrix} -1/2 \\ 1 \\ -1/2 \end{pmatrix} = \sqrt{\frac{2}{3}} \cdot \begin{pmatrix} -1/2 \\ 1 \\ -1/2 \end{pmatrix} = \begin{pmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{pmatrix}$

$q_3 = \frac{1}{\|v_2\|} \cdot v_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} \Rightarrow$

$A = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 0 & 0 & \\ 0 & 0 & \\ & & 3 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix}^t$

(2.) (a) $B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} = 2A \Rightarrow$

$B = 2 \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix}^t =$

$= Q \cdot \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 6 \end{pmatrix} \cdot Q^t$

(b) $B = A - 3I_3 = Q \Delta Q^t - 3 \underbrace{Q Q^t}_{= I_3} = Q(\Delta - 3I_3) Q^t$

$= Q \cdot \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot Q^t$

(c) $B = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix} = \frac{1}{2} A - \frac{1}{2} I_3 = \frac{1}{2}(A - I_3) =$

$= \frac{1}{2} (Q \Delta Q^t - Q Q^t) = Q \left(\frac{1}{2} \Delta - \frac{1}{2} I_3 \right) Q^t = Q \cdot \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot Q^t$

(3.) A matrice simetrică are toate valorile proprii $\in \mathbb{R}^n$ strict pozitive $\Leftrightarrow \langle u, v \rangle = u^t A v$ este produs scalar pe

Spaţiu euclidian

Definiţie 7.2. Fie V un $\mathbb{R}$-spaţiu vectorial. Un **produs scalar** pe V este o funcţie

$$\langle\ ,\ \rangle : V \times V \to \mathbb{R} \qquad \mathbf{u}, \mathbf{v} \in V \mapsto \langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}$$

astfel încât

1) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$

2) $\langle \mathbf{u}_1 + \mathbf{u}_2, \mathbf{v} \rangle = \langle \mathbf{u}_1, \mathbf{v} \rangle + \langle \mathbf{u}_2, \mathbf{v} \rangle$

3) $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle$

4) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$

5) $\langle \mathbf{u}, \mathbf{u} \rangle = 0 \iff \mathbf{u} = \mathbf{0}$

Un spaţiu vectorial real finit dimensional V înzestrat cu un produs scalar $\langle\ ,\ \rangle$ se numeşte **spaţiu euclidian** $(V, \langle\ ,\ \rangle)$.

„$\Rightarrow$" 1) $\langle u, v \rangle = u^t A v = (u^t A v)^t = v^t A (u^t)^t = v^t A u =$

$= \langle v, u \rangle$

2) $\langle u_1 + u_2, v \rangle = (u_1 + u_2)^t A v = (u_1^t + u_2^t) A v = u_1^t A v + u_2^t A v$

$= \langle u_1, v \rangle + \langle u_2, v \rangle$

3) $\langle \lambda u, v \rangle = (\lambda u)^t A v = \lambda u^t A v = \lambda \langle u, v \rangle$

4) $\langle u, u \rangle = u^t A u = \underbrace{u^t Q}_{v^t} \Lambda \underbrace{Q^t u}_{v} = v^t \Lambda v$

Notăm $v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \Rightarrow \langle u, u \rangle = (x_1\ x_2\ \dots\ x_n) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} =$

$= x_1^2 \lambda_1 + x_2^2 \lambda_2 + \dots + x_n^2 \lambda_n \geq 0$, deoarece valorile proprii ale matricei simetrice A sunt strict pozitive

5) $\langle u, u \rangle = 0 \overset{4)}{\iff} x_1^2 \lambda_1 + \dots + x_n^2 \lambda_n = 0 \overset{\text{ipoteză}}{\iff} x_1 = \dots = x_n = 0$

$\iff \boxed{u = 0}$

„$\Leftarrow$" $A = (a_{ij})_{i=\overline{1,n}}$ și $\{e_1, \dots, e_n\}$ bază canonică din $\mathbb{R}^n$

$\Rightarrow a_{ij} = \underset{\text{prod. scalar}}{\langle e_i, e_j \rangle} = \langle e_j, e_i \rangle = a_{ji} \Rightarrow$ A este matrice simetrică

$\rightarrow$ $A = Q \Lambda Q^t$

$u_i = Q e_i \neq 0$ (vectorii $g_1, \dots, g_n$ $Q = (g_1 | g_2 | \dots | g_n)$ sunt ortogonali)

$\Rightarrow \langle u_i, u_i \rangle > 0$

$\langle u_i, u_i \rangle = u_i^t A u_i = (Q e_i)^t Q \Lambda \underbrace{Q^t Q}_{I_n} e_i = e_i^t \underbrace{Q^t Q}_{I_n} \Lambda e_i =$

$= e_i^t A e_i = \lambda_i$. Cum $\langle u_i, u_i \rangle > 0 \Rightarrow \lambda_i > 0 \Rightarrow$
Valorile proprii ale matricii A sunt strict pozitive.

(4.) $A \in \mathcal{M}_n(\mathbb{R})$ matrice simetrică
cu toate valorile proprii strict pozitive

$Ax = b \Rightarrow x$ minimizează
funcția $f: \mathbb{R}^n \to \mathbb{R}^n$,
$f(x_1, \dots, x_n) = \sum_{i,j=1}^{n} a_{ij} x_i x_j - 2\sum_{i=1}^{n} b_i x_i$

Soluție
Determinăm punctele de extreme ale funcției f și demonstrăm că x este punct de minim
Determinăm punctele critice ale funcției f
(printre ele se regăsesc și punctele de extreme)

$\frac{\partial f}{\partial x_i}(x_1, \dots, x_n) = 0, \quad i = \overline{1,n} \Leftrightarrow \sum_{j=1}^{n} \widehat{a_{ij}} \cdot x_j +$
$\sum_{j=1}^{n} \widehat{a_{ji}} \cdot x_j = 2b_i \overset{\text{A sim.}}{\Leftrightarrow} \sum_{j=1}^{n} a_{ij} \cdot x_j = b_i \quad i = \overline{1,n} \Leftrightarrow Ax = b$

$\Rightarrow$ Punctele critice sunt soluțiile ecuației $Ax = b$
Determinăm matricea hessiană

$H = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{i,j = \overline{1,n}}$

$\frac{\partial f}{\partial x_i} = 2 \sum_{j=1}^{n} a_{ij} \cdot x_j - 2b_i$

$\frac{\partial f}{\partial x_i \partial x_j} = 2a_{ij} \Rightarrow H = 2A \Rightarrow$ valorile proprii ale lui H sunt strict pozitive $\Rightarrow$ soluția sistemului $Ax = b$ este punct extrem de minim $\Rightarrow x$ minimizează funcția f.

(5) Inegalitatea Cauchy-Schwartz generalizată

A matrice simetrică cu val. proprii pozitive

$$|\langle Ax, y\rangle| \leq \sqrt{\langle Ax, x\rangle} \cdot \sqrt{\langle Ay, y\rangle}$$

Soluție

$A = Q \Delta Q^t$, $\Delta = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$, $\lambda_1, \dots, \lambda_n \geq 0$

Fie $u \in \mathbb{R}^n$, $u = \begin{pmatrix} u_1 \\ u_n \end{pmatrix}$

$$\langle Au, u\rangle = (Au)^t \cdot u = u^t A u = \overbrace{u^t Q}^{v^t} \Delta \overbrace{Q^t u}^{v} =$$

$$= v^t \Delta v = v_1^2 \cdot \lambda_1 + v_2^2 \cdot \lambda_2 + \dots + v_n^2 \cdot \lambda_n \geq 0,$$ deoarece valorile proprii sunt pozitive. (1)

Notăm $z_\lambda = x + \lambda \langle Ax, y\rangle y$, $x, y \in \mathbb{R}^n$, $\lambda \in \mathbb{R} \Rightarrow$

Din (1) $\rightarrow$ $\langle Az_\lambda, z_\lambda\rangle \geq 0 \iff$

$$\langle A(x + \lambda\langle Ax, y\rangle y),\ x + \lambda\langle Ax, y\rangle y\rangle \geq 0 \iff$$

$$\langle Ax, x + \lambda\langle Ax, y\rangle y\rangle + \langle A\lambda\langle Ax, y\rangle y,\ x + \lambda\langle Ax, y\rangle y\rangle$$

$$\geq 0 \iff \langle Ax, x\rangle + \langle Ax, \lambda\langle Ax, y\rangle y\rangle +$$

$$\langle A\lambda\langle Ax, y\rangle y,\ x\rangle + \langle A\lambda\langle Ax, y\rangle y,\ \lambda\langle Ax, y\rangle y\rangle \geq 0$$

$$\iff \langle Ax, x\rangle + \lambda\langle Ax, y\rangle\langle Ax, y\rangle +$$

$$\lambda\underbrace{\langle Ax, y\rangle}_{=}\langle Ay, x\rangle + \lambda^2\langle Ax, y\rangle^2\langle Ay, y\rangle \geq 0 \iff$$

$$\langle Ax, y\rangle^2\langle Ay, y\rangle\lambda^2 + 2\lambda\langle Ax, y\rangle^2 + \langle Ax, x\rangle \geq 0$$

$\Rightarrow$ Vârful parabolei se află deasupra axei $Ox \Rightarrow$

$$\frac{-\Delta}{4a} \geq 0$$

$$\Delta = 4\langle Ax, y\rangle^4 - 4\langle Ax, x\rangle\langle Ax, y\rangle^2\langle Ay, y\rangle \Rightarrow$$

$$\frac{-\Delta}{4a} = \frac{-(4\langle Ax,y\rangle^2 - 4\langle Ax,x\rangle\cdot\langle Ay,y\rangle)}{4\langle Ax,y\rangle^2\langle Ay,y\rangle} \geq 0 \iff$$

$$\langle Ax,y\rangle^2 - \langle Ax,x\rangle\langle Ay,y\rangle \leq 0 \iff \langle Ax,y\rangle^2 \leq \langle Ax,x\rangle\cdot\langle Ay,y\rangle$$

$$\iff \boxed{|\langle Ax,y\rangle| \leq \sqrt{\langle Ax,x\rangle}\cdot\sqrt{\langle Ay,y\rangle}}$$

(6)

(a) $g \in \mathbb{R}^n$, $\|g\| = 1$, $W = Sp\{g\}$

$$pr_W^v = pr_g^v = \frac{\langle v,g\rangle}{\langle g,g\rangle}\cdot g = \langle v,g\rangle\cdot g = g\cdot\langle v,g\rangle =$$

$$= g g^t v \quad (\langle v,g\rangle = g^t v = v^t g)$$

(b) $f: \mathbb{R}^n \to \mathbb{R}^n$

$f(v) = pr_W^v$

Aplicație liniară

1) $f(x+y) = pr_W^{(x+y)} = pr_g^{(x+y)} = \frac{\langle (x+y),g\rangle}{\langle g,g\rangle}\cdot g =$

$$= \frac{\langle x,g\rangle + \langle y,g\rangle}{\langle g,g\rangle}\cdot g = pr_g^x + pr_g^y = f(x) + f(y)$$

2) $f(\lambda x) = pr_W^{\lambda x} = pr_g^{\lambda x} = \frac{\langle \lambda x,g\rangle}{\langle g,g\rangle}\cdot g = \frac{\lambda\langle x,g\rangle}{\langle g,g\rangle}\cdot g =$

$$= \lambda pr_g^x = \lambda f(x)$$

Din 1), 2) $\Rightarrow$ f aplicație liniară $\{e_1,\dots,e_n\}$ → bază canonică

$$f(e_1) = pr_W^{e_1} = \frac{\langle e_1,g\rangle}{\langle g,g\rangle}\cdot g = g_1\cdot g$$

$$\vdots$$

$$f(e_n) = \dots = g_n\cdot g$$

Imaginea este de dim. 1 $\Rightarrow$ Rangul matricii este 1

$$\Rightarrow A = (\underbrace{g_1 g \mid g_2 g \mid \dots \mid g_n g}) = \boxed{g\cdot g^t}$$

(c) $g_1, \dots, g_k \in \mathbb{R}^n$, $W = Sp\{g_1, \dots, g_n\}$

$pr_W^v = pr_{g_1}^v + \dots + pr_{g_n}^v \overset{①}{=} (g_1 g_1^t) + \dots + (g_k g_k^t)$

$f(e_1) = pr_W e_1 = g_{11} \cdot g_1 + \dots + g_{n1} g_n$

$\vdots$

$f(e_n) = pr_W e_n = g_{1n} g_1 + \dots + g_{nn} g_n$

$\Rightarrow A = (g_{11} g_1 + \dots + g_{n1} g_n | \dots | g_{1n} g_1 + \dots + g_{nn} g_n) =$

$= (g_1 g_1^t) + \dots + (g_n g_n^t)$

(d) $A^t = (g_1 g_1^t + \dots + g_n g_n^t)^t = (g_1 g_1^t)^t + \dots + (g_n g_n^t)^t =$

$= (g_1^t)^t g_1^t + \dots + (g_n^t)^t g_n^t = g_1 g_1^t + \dots + g_n g_n^t = A \Rightarrow$

A este o <u>matrice simetrică</u>.

$A^2 = \left(\sum_{i=1}^n g_i g_i^t\right)^2 = \sum_{i,j=1}^n g_i g_i^t g_j g_j^t = \sum_{i=1}^n g_i g_i^t = A$

$g_i^t g_j = \begin{cases} 0, i \neq j & g_i, g_j \text{ ortogonali (ortonormați)} \\ 1, i = j & \text{(versor)} \end{cases}$

(e) $Q = (g_1 | \dots | g_k)$ → ortonormați și deci liniar independenți

$A = g_1 g_1^t + \dots + g_n g_n^t = QQ^t$

$rang\, Q = k \Rightarrow$

$\boxed{rang\, A = rang(QQ^t) = rang\, Q = k}$

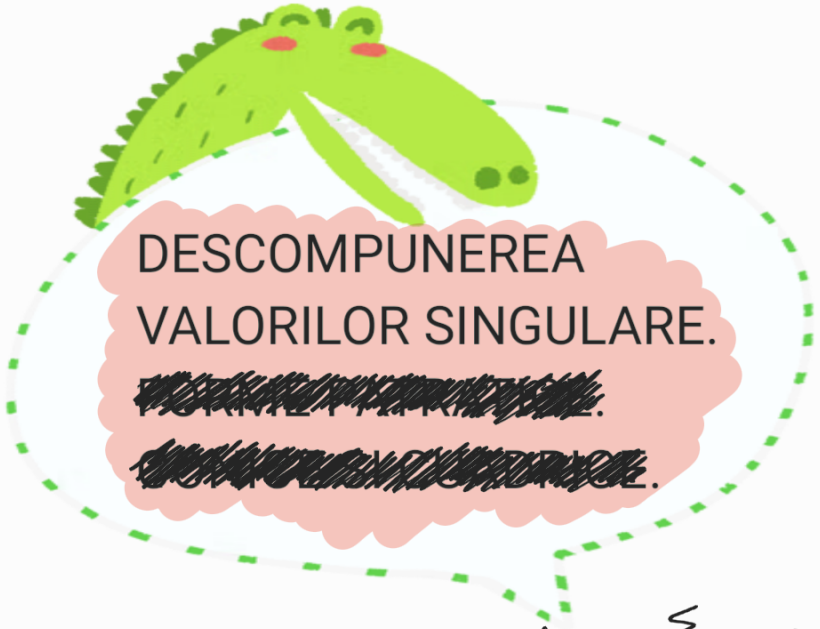
(f) Completăm la o bază ortonormată în $\mathbb{R}^n$

$Q_* = (g_1 | \dots | g_k | g_{k+1} | \dots | g_n)$ → matrice ortogonală

$A g_i = (g_1 g_1^t + \dots + g_n g_n^t) g_i = g_1 g_1^t g_i + \dots +$ (produs scalar $= 0$)

$g_n g_n^t g_i = 0 \quad (\forall) \; i = \overline{k+1, n} \Rightarrow$

$$\boxed{A = Q_* \cdot \left(\begin{array}{c|c} 1 & \\ \ddots & 0 \\ 1 & \\ \hline 0 & 0 \end{array}\right) Q_*^t}$$



1.

(a) $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$ $(3,2) = \underbrace{(3,3)}_{U}\underbrace{(3,2)}_{\Sigma}\underbrace{(2,2)}_{V}$

$\boxed{A = U\Sigma V^t}$

$A^tA = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$

$\det(A - \lambda I_2) = -\lambda(4-\lambda) \Rightarrow \lambda_1 = 4$ și $\lambda_2 = 0$

Pentru $\lambda_1 = 4 \Rightarrow \begin{cases} 4y = 4y \Rightarrow V_{\lambda_1} = Sp\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}\} \\ 4x = 0 \end{cases}$ } vectori ortonormați

Pentru $\lambda_2 = 0 \Rightarrow 4y = 0 \Rightarrow V_{\lambda_2} = Sp\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\}$

$\Rightarrow V = (v_1 | v_2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\sigma_1 = \sqrt{\lambda_1} = 2, \ \sigma_2 = 0 \Rightarrow \Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

$u_1 = \frac{1}{\sigma_1} \cdot A v_1 = \frac{1}{2}\begin{pmatrix} 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{2} \cdot \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Prelungim la o bază ortonormată $\{u_1, u_2, u_3\}$

$= \{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\} \Rightarrow U = (u_1 | u_2 | u_3) = I_3 \Rightarrow$

$\boxed{A = U\Sigma V^t = I_3 \cdot \begin{pmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^t}$

(b) $A=\begin{pmatrix}1&1\\1&1\end{pmatrix} = \underset{(2,2)}{U}\underset{(2,2)}{\Sigma}\underset{(2,2)}{V^t}$

$$A^tA = \begin{pmatrix}1&1\\1&1\end{pmatrix}\begin{pmatrix}1&1\\1&1\end{pmatrix} = \begin{pmatrix}2&2\\2&2\end{pmatrix} = 2\begin{pmatrix}1&1\\1&1\end{pmatrix}$$

$\det(A-\lambda_2 I_2) = (1-\lambda)^2 - 1 = \lambda^2 - 2\lambda + 1 - 1 = \lambda^2 - 2\lambda = \lambda(\lambda-2)$

$\Rightarrow$ $\boxed{\lambda_1 = 2}$ și $\boxed{\lambda_2 = 0}$

Pentru $\lambda_1 = 2 \Rightarrow \begin{cases} x+y = 2x \\ x+y = 2y \end{cases} \Rightarrow x+y=0 \Rightarrow V_{\lambda_1} = Sp\{\begin{pmatrix}1\\1\end{pmatrix}\}$

$v_1 = \begin{pmatrix}1\\1\end{pmatrix}$

$g_1 = \frac{1}{\|v_1\|}\cdot v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$

Pentru $\lambda_2 = 0 \Rightarrow x+y=0 \Rightarrow x=-y \Rightarrow V_{\lambda_2} = Sp\{\begin{pmatrix}-1\\1\end{pmatrix}\}$

$g_2 = \frac{1}{\|v_2\|}\cdot v_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix} = \begin{pmatrix}-1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$

$\Rightarrow \boxed{V = (g_1|g_2) = \begin{pmatrix}1/\sqrt{2} & -1/\sqrt{2}\\ 1/\sqrt{2} & 1/\sqrt{2}\end{pmatrix}}$

$\sigma_1 = \sqrt{\lambda_1} = \sqrt{2} \Rightarrow$

$\boxed{\Sigma = \begin{pmatrix}\sqrt{2} & 0\\ 0 & 0\end{pmatrix}}$

$u_1 = \frac{1}{\sigma_1}\cdot Ag_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&1\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}2/\sqrt{2}\\2/\sqrt{2}\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}$

$w_1 = \frac{1}{\|u_1\|}\cdot u_1 = \frac{1}{\sqrt{2}}\cdot\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$

Prelungim la o bază ortonormată $\{w_1, w_2, w_3\}$

$w_1,\ a_2 = \begin{pmatrix}0\\1\end{pmatrix}$

$u_2 = a_2 - pr_{w_1}^{a_2} = \begin{pmatrix}0\\1\end{pmatrix} - \frac{\langle a_2, w_1\rangle}{\langle w_1, w_1\rangle}\cdot w_1 =$

$= \begin{pmatrix}0\\1\end{pmatrix} - \frac{1}{\sqrt{2}}\cdot\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix} - \begin{pmatrix}1/2\\1/2\end{pmatrix} = \begin{pmatrix}-1/2\\1/2\end{pmatrix}$

$w_2 = \frac{1}{\|u_2\|}\cdot\begin{pmatrix}-1/2\\1/2\end{pmatrix} = \frac{1}{\sqrt{\frac{1}{4}+\frac{1}{4}}}\cdot\begin{pmatrix}-1/2\\1/2\end{pmatrix} = \sqrt{2}\begin{pmatrix}-1/2\\1/2\end{pmatrix} = \begin{pmatrix}-1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$

$\Rightarrow \boxed{U = (w_1|w_2) = \begin{pmatrix}1/\sqrt{2} & -1/\sqrt{2}\\ 1/\sqrt{2} & 1/\sqrt{2}\end{pmatrix}}$

(C) $A = \begin{pmatrix} -2 & 11 \\ -10 & 5 \end{pmatrix}$ $\quad A = U\Sigma V^t$ $\quad (2,2)\ (2,2)\ (2,2)$

$$A^tA = \begin{pmatrix} -2 & -10 \\ 11 & 5 \end{pmatrix}\begin{pmatrix} -2 & 11 \\ -10 & 5 \end{pmatrix} = \begin{pmatrix} 104 & -72 \\ -72 & 146 \end{pmatrix}$$

$P_A(\lambda) = \det(A - \lambda I_2) = (104-\lambda)(146-\lambda) - 72^2 = 15184 - 104\lambda - 146\lambda + \lambda^2 - 72^2 = \lambda^2 - 250\lambda + 10000$

$\Delta = 22500 \quad \lambda_{1,2} = \frac{250 \pm 150}{2}$ (50, 200) $\Rightarrow$

$\lambda_1 = 200, \lambda_2 = 50$

Pentru $\lambda_1 = 200 \Rightarrow$

$$\begin{cases} 104x - 72y = 200x \\ -72x + 146y = 200y \end{cases} \Leftrightarrow 96x = -72y \Leftrightarrow 4x = -3y \Rightarrow V_{\lambda_1} = Sp\left\{\begin{pmatrix} -3 \\ 4 \end{pmatrix}\right\}$$

Pentru $\lambda_2 = 50 \Rightarrow$

$$\begin{cases} 104x - 72y = 50x \\ -72x + 146y = 50y \end{cases} \Leftrightarrow 54x = 72y \Leftrightarrow 3x = 4y \rightarrow V_{\lambda_2} = Sp\left\{\begin{pmatrix} 4 \\ 3 \end{pmatrix}\right\}$$

$v_1 = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \quad g_1 = \frac{1}{\|v_1\|} \cdot v_1 = \frac{1}{\sqrt{9+16}} \begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{pmatrix}$

$v_2 = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \quad g_2 = \frac{1}{\|v_2\|} v_2 = \frac{1}{\sqrt{16+9}} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}$

$\Rightarrow V = (v_1 | v_2) = \begin{pmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{pmatrix}$

$\sqrt{1} = 10\sqrt{2}, \sqrt{2} = 5\sqrt{2} \Rightarrow$

$$\Sigma = \begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 5\sqrt{2} \end{pmatrix}$$

$$u_1 = \frac{1}{\sqrt{1}} \cdot Ag_1 = \frac{1}{10\sqrt{2}} \cdot \begin{pmatrix} -2 & 11 \\ -10 & 5 \end{pmatrix}\begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix} = \frac{1}{10\sqrt{2}} \begin{pmatrix} 6/5 + 44/5 \\ 30/5 + 20/5 \end{pmatrix}$$

$$= \frac{1}{10\sqrt{2}} \begin{pmatrix} 10 \\ 10 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$u_2 = \frac{1}{\sqrt{2}} \cdot Ag_2 = \frac{1}{5\sqrt{2}} \begin{pmatrix} -2 & 11 \\ -10 & 5 \end{pmatrix}\begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix} = \frac{1}{5\sqrt{2}} \begin{pmatrix} -8/5 + 33/5 \\ -40/5 + 15/5 \end{pmatrix} =$$

$$= \frac{1}{5\sqrt{2}} \cdot \begin{pmatrix} 5 \\ -5 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

$$\Rightarrow U = (u_1 | u_2) = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

(d) $A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$ $\quad A = \underset{(2,2)}{U}\underset{(2,3)}{\Sigma}\underset{(3,3)}{V^t}$

$$A^tA = \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} = \begin{pmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{pmatrix}$$

$$P_A(\lambda) = \begin{vmatrix} 13-\lambda & 12 & 2 \\ 12 & 13-\lambda & -2 \\ 2 & -2 & 8-\lambda \end{vmatrix} = -\lambda^3 + 34\lambda^2 - 225\lambda =$$

$$= \lambda(-\lambda^2 + 34\lambda - 225) =$$

$$= \lambda(\lambda-9)(\lambda-25)$$

Pentru $\lambda_1 = 25 \Rightarrow \begin{cases} 13x + 12y + 2z = 25x \\ 12x + 13y - 2z = 25y \\ 2x - 2y + 8z = 25z \end{cases} \Leftrightarrow z = \frac{12x - 12y}{2} = 6x - 6y$

$\Rightarrow \quad 2x - 2y + 48x - 48y = 150x - 150y \Leftrightarrow 100z = 100y \Leftrightarrow x = y \Rightarrow$

$$V_{\lambda_1} = Sp\left\{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right\}$$

Pentru $\lambda_2 = 9 \Rightarrow \begin{cases} 13x + 12y + 2z = 9x \\ 12x + 13y - 2z = 9y \\ 2x - 2y + 8z = 9z \end{cases} \Rightarrow V_{\lambda_2} = Sp\left\{\begin{pmatrix} 1/4 \\ -1/4 \\ 1 \end{pmatrix}\right\}$

Pentru $\lambda_3 = 0 \Rightarrow \begin{cases} 13x + 12y + 2z = 0 \\ 12x + 13y - 2z = 0 \\ 2x - 2y + 8z = 0 \end{cases} \Rightarrow V_{\lambda_3} = Sp\left\{\begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}\right\}$

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \Rightarrow \quad g_1 = \frac{1}{\|v_1\|}\cdot\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} 1/4 \\ -1/4 \\ 1 \end{pmatrix} \Rightarrow \quad g_2 = \frac{1}{\|v_2\|}\begin{pmatrix} 1/4 \\ -1/4 \\ 1 \end{pmatrix} = \frac{2\sqrt{2}}{3}\cdot\begin{pmatrix} 1/4 \\ -1/4 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2}\cdot 3 \\ -1/3\sqrt{2} \\ 4/3\sqrt{2} \end{pmatrix}$$

$$v_3 = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} \Rightarrow \quad g_3 = \frac{1}{\|v_3\|}\cdot\begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2/3 \\ 2/3 \\ 1/3 \end{pmatrix} \Rightarrow$$

$$V = (v_1|v_2|v_3) = \begin{pmatrix} 1/\sqrt{2} & 1/3\sqrt{2} & -2/3 \\ 1/\sqrt{2} & -1/3\sqrt{2} & 2/3 \\ 0 & 4/3\sqrt{2} & 1/3 \end{pmatrix}$$

$\sqrt{\lambda_1} = 5, \ \sqrt{\lambda_2} = 3 \Rightarrow$

$$\Sigma = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$$

$u_1 = \frac{1}{\sigma_1} \cdot Ag_1 = \frac{1}{5} \cdot \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 5/\sqrt{2} \\ 5/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

$u_2 = \frac{1}{\sigma_2} \cdot Ag_2 = \frac{1}{3} \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} \begin{pmatrix} 1/3\sqrt{2} \\ -1/3\sqrt{2} \\ 4/3\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$

$\Rightarrow \boxed{U = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}}$

(e) $A = \begin{pmatrix} 4 & -2 \\ 2 & -1 \\ 0 & 0 \end{pmatrix}$ $\quad A = \underbrace{U \cdot \Sigma \cdot V^t}_{(3,3)(3,2)(2,2)}$

$A^t A = \begin{pmatrix} 4 & 2 & 0 \\ -2 & -1 & 0 \end{pmatrix} \begin{pmatrix} 4 & -2 \\ 2 & -1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 20 & -10 \\ -10 & 5 \end{pmatrix} = 5 \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}$

$P_A(\lambda) = \begin{vmatrix} 20-\lambda & -10 \\ -10 & 5-\lambda \end{vmatrix} = \lambda^2 - 25\lambda = \lambda(\lambda - 25) \Rightarrow \lambda_1 = 25, \lambda_2 = 0$

Pentru $\lambda = 25 \Rightarrow \begin{cases} 20x - 10y = 25x \\ -10x + 5y = 25y \end{cases} \Rightarrow x = -2y \Rightarrow V_{\lambda_1} = Sp\{\begin{pmatrix} -2 \\ 1 \end{pmatrix}\}$

Pentru $\lambda = 0 \Rightarrow \begin{cases} 20x - 10y = 0 \\ -10x + 5y = 0 \end{cases} \rightarrow y = 2x \Rightarrow V_{\lambda_2} = Sp\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\}$

$v_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \Rightarrow q_1 = \frac{1}{\|v_1\|} \cdot v_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix}$

$v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \Rightarrow q_2 = \frac{1}{\|v_2\|} v_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{pmatrix}$

$\Rightarrow \boxed{V = \begin{pmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix}} \quad \sigma_1 = 5$

$\boxed{\Sigma = \begin{pmatrix} 5 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}}$ $\quad u_1 = \frac{1}{\sigma_1} \cdot Ag_1 = \frac{1}{5} \begin{pmatrix} 4 & -2 \\ 2 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix} =$

$= \begin{pmatrix} -2/\sqrt{5} \\ -1/\sqrt{5} \\ 0 \end{pmatrix}$ Prelungim la o bază ortonormată $\{u_1, u_2, u_3\}$ vectorii $u_1, e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\Rightarrow w_2 = e_1 - pr_{u_1} e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{\langle e_1, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \frac{2}{\sqrt{5}} \begin{pmatrix} -2/\sqrt{5} \\ -1/\sqrt{5} \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 1/5 \\ -2/5 \\ 0 \end{pmatrix}$ $\quad u_2 = \frac{1}{\|w_2\|} w_2 = \sqrt{5} \begin{pmatrix} 1/5 \\ -2/5 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \\ 0 \end{pmatrix}$

$$w_3 = e_3 - \underset{u_1}{pr} e_3 - \underset{u_2}{pr} e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \underbrace{\frac{\langle u_1, e_3 \rangle}{\langle u_1, u_1 \rangle}}_{1} \cdot u_1 - \underbrace{\frac{\langle u_2, e_3 \rangle}{\langle u_2, u_2 \rangle}}_{1} \cdot u_2 \in \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \quad U = \begin{pmatrix} -2/\sqrt{5} & 1/\sqrt{5} & 0 \\ -1/\sqrt{5} & 2/\sqrt{5} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$